Drones and Unmanned Aerial Vehicles

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Introduction to Drones

A drone, formally called an Unmanned Aerial Vehicle (UAV) or Unmanned Aircraft System (UAS), is an aircraft that flies without a human pilot on board. It is controlled remotely by a pilot on the ground using a transmitter or a smartphone, or it flies automatically along a pre-programmed route using GPS and onboard computers. Drones range from tiny toy models that weigh a few grams to large military aircraft weighing several tonnes. Today they are used for photography, delivery, farming, surveying, mapping, disaster relief, inspection and defence.

History

Early unmanned aircraft were developed for military use, including target drones in the early twentieth century. The word "drone" was inspired by the male honey bee. Modern military UAVs such as the American Predator entered service in the 1990s. Commercial and hobby drones became popular after 2010 with the release of consumer quadcopters like the Parrot AR.Drone and, from 2013, the DJI Phantom, because of cheap sensors, lithium batteries and powerful small chips.

How Does a Quadcopter Stay in the Air?

A quadcopter is a drone with four rotors, each with a propeller driven by an electric motor. A quadcopter stays in the air because its propellers push air downward, and by Newton's third law the air pushes the drone upward. This upward force is called thrust or lift. When the total thrust equals the weight of the drone, it hovers. To rise, all four motors speed up together; to descend, they slow down together.

The four rotors do not all spin the same way. Two spin clockwise and two spin anticlockwise, with opposite pairs matched, so that the twisting forces (torque) cancel out and the drone does not spin uncontrollably. To move forward, the rear motors speed up and the front motors slow down, tilting the drone nose-down; this is called pitch. To move sideways it changes the speeds of left and right motors, called roll. To turn left or right (yaw), it speeds up one diagonal pair of motors relative to the other, creating a net torque. A flight controller, a small computer with an IMU containing gyroscope and accelerometer, a barometer for height, a compass and GPS, adjusts motor speeds hundreds of times every second to keep the drone stable. Other drone designs include fixed-wing drones, which fly like aeroplanes and are efficient for long distances, and hybrid vertical take-off and landing (VTOL) drones.

Main Components

A drone consists of a frame, motors, electronic speed controllers (ESCs), propellers, a lithium-polymer battery, the flight controller, a radio receiver and optional payloads such as a camera, a gimbal, sprayers or sensors. Typical consumer drones fly for 20 to 40 minutes on one battery charge.

Uses of Drones in Agriculture

Two important uses of drones in agriculture are spraying and crop monitoring. Agricultural spraying drones carry tanks of pesticide, herbicide or liquid fertiliser and cover an acre in about ten minutes, which is much faster than manual spraying and reduces farmers' exposure to chemicals and saves water. Crop monitoring drones carry multispectral or thermal cameras that capture images of fields and identify plant stress, pest attacks, water shortage and nutrient deficiency using indices such as NDVI. Other uses include field mapping and surveying, seed sowing, livestock monitoring and estimating crop yield and damage for insurance claims.

Other Applications

Drones deliver medicines and blood to remote areas, inspect power lines, bridges and wind turbines, survey land and mines, monitor traffic, support search and rescue after floods and earthquakes, film movies and sports, and perform light shows. Military drones are used for surveillance and strikes.

What Rules Apply to Flying Drones in India?

India regulates drones through the Drone Rules, 2021, notified by the Ministry of Civil Aviation in August 2021, which liberalised earlier rules. Drones must be registered and flown through the Digital Sky Platform. The airspace is divided into green zones (flying allowed without permission up to 400 feet or 120 metres), yellow zones (permission needed) and red zones (flying prohibited, such as near airports, international borders and sensitive government buildings). Drones are classified by weight: nano (up to 250 grams), micro (above 250 grams up to 2 kg), small (2 to 25 kg), medium (25 to 150 kg) and large (above 150 kg). Nano drones and micro drones for non-commercial use do not need a pilot licence, while others need a remote pilot certificate from a DGCA-approved drone school. Drones must be flown in the visual line of sight, not in no-fly zones, and not over crowds. Operators must follow privacy and safety rules, and heavy penalties apply for violations. Import of foreign-made drones was banned in 2022 to promote domestic manufacturing, and a production-linked incentive scheme supports Indian makers.

Drones in India

Chennai-based Garuda Aerospace manufactures agricultural and surveillance drones, and several startups in Tamil Nadu and Bengaluru build drones for defence and farming. The Namo Drone Didi scheme, approved in 2023, trains women self-help groups to operate agricultural drones and provides them with drones for spraying. ISRO and the Survey of India use drones for mapping under the SVAMITVA scheme, which maps village property. Drones carried medicines to remote areas of Telangana and the Northeast in trials.

Challenges and Future

Challenges include short battery life, safety, privacy, airspace congestion and misuse. The future includes drone delivery, urban air taxis, swarm drones guided by AI, and better counter-drone systems.

Quick Facts

- A quadcopter stays up because its four propellers push air downward and create thrust; two rotors spin clockwise and two anticlockwise to cancel torque.
- Drones in agriculture are used for spraying pesticides and fertiliser and for crop health monitoring.
- India's Drone Rules, 2021 divide airspace into green, yellow and red zones and use the Digital Sky Platform.
- Indian drones are classed as nano, micro, small, medium and large by weight.
- The Namo Drone Didi scheme gives agricultural drones to women self-help groups.